

PSYCHOLOGY

SECTION I

Time—1 hour and 10 minutes

100 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding oval on the answer sheet.

1. The longer an individual is exposed to a strong odor, the less aware of the odor the individual becomes. This phenomenon is known as sensory
(A) acuity
(B) adaptation
(C) awareness
(D) reception
(E) overload
2. Which type of psychologist would be interested primarily in studying whether people behave differently in groups than they do when alone?
(A) Experimental
(B) Cognitive
(C) Developmental
(D) Social
(E) Clinical
3. The basic purpose of the *DSM-IV-TR* is to
(A) provide a set of diagnostic categories for classifying psychological disorders
(B) provide a clear distinction between neurosis and psychosis
(C) describe the psychoanalytic approach to psychological disorders
(D) describe internal personality factors that are involved in psychological disorders
(E) identify childhood experiences that contribute to psychological disorders
4. An individual survives a period of captivity and exhibits behaviors that include anxiety, inability to concentrate, depression, edginess, and the reexperience of stressful events. These symptoms illustrate which of the following disorders?
(A) Generalized anxiety
(B) Major depression
(C) Hypochondriasis
(D) Histrionic
(E) Posttraumatic stress
5. Which of the following theoretical frameworks would argue most strongly that a healthy child will choose what is good for his or her growth?
(A) Social learning theory
(B) Psychoanalytic
(C) Behavioral
(D) Humanistic
(E) Psychodynamic
6. The ability to choose specific stimuli to learn about, while filtering out or ignoring other information, is called
(A) selective attention
(B) subliminal perception
(C) time-sharing
(D) masking
(E) shadowing

7. A two-year-old child is frightened by a small dog. A few weeks later the same child sees a cat and becomes frightened. The child's reaction is most likely an example of which of the following?
- (A) Stimulus discrimination
 - (B) Second-order conditioning
 - (C) Stimulus generalization
 - (D) Sensory preconditioning
 - (E) Spontaneous recovery
8. A researcher asks elementary, junior high, senior high, and college students to define the term "cheating," and analyzes differences in their definitions across age groups. This is an example of which type of study?
- (A) Longitudinal
 - (B) Sequential
 - (C) Cross-sectional
 - (D) Case study
 - (E) Observational
9. Rudolph spends hours painting in his studio, even though he sells few pictures. Which of the following explains Rudolph's creative productivity?
- (A) Functional fixedness
 - (B) Inductive reasoning
 - (C) Intrinsic motivation
 - (D) Incubation
 - (E) Heuristics

Section I

Questions 10-12 refer to the following definitions. Each definition can be used once, more than once, or not at all.

- (A) Prototype matching to organize information into categories
 - (B) Maintaining information in memory through repetition
 - (C) Differential treatment, usually negative, based on group membership
 - (D) Recognizing an object as distinct from its surroundings
 - (E) Learning to respond differently to similar stimuli
10. Which is a definition of discrimination that most directly applies to perception?
- (A) A
 - (B) B
 - (C) C
 - (D) D
 - (E) E
11. Which is a definition of discrimination that most directly applies to classical conditioning?
- (A) A
 - (B) B
 - (C) C
 - (D) D
 - (E) E
12. Which is a definition of discrimination that most directly applies to social psychology?
- (A) A
 - (B) B
 - (C) C
 - (D) D
 - (E) E
13. Contemporary definitions of abnormality might include all of the following criteria EXCEPT
- (A) maladaptiveness
 - (B) personal distress or discomfort
 - (C) deviance from cultural norms
 - (D) gender
 - (E) statistical prevalence
14. The correlations between the IQ scores of identical twins reared apart are lower than those of identical twins reared together. This difference is best explained by which of the following?
- (A) Heredity plays an important role in determining IQ.
 - (B) Environment plays an important role in determining IQ.
 - (C) Heredity plays no role in determining IQ.
 - (D) Environment plays no role in determining IQ.
 - (E) Heredity and environment play an equal role in determining IQ.
15. A test that fails to predict what it is designed to predict lacks
- (A) standardization
 - (B) norms
 - (C) fairness
 - (D) validity
 - (E) reliability
16. Neurosurgeons cut the corpus callosum in the brain disrupting communication between the right and left hemispheres to
- (A) prevent the spread of epileptic seizures
 - (B) reduce anxiety attacks and phobic reactions
 - (C) reduce the incidence of violent behaviors
 - (D) treat schizophrenia
 - (E) reduce mood swings

17. Keisha was able to roll over at two months, crawl at five months, and walk at ten months. This sequence of development is most likely due to
- (A) egocentrism
 - (B) social context
 - (C) maturation
 - (D) attachment
 - (E) assimilation
18. A psychologist is attempting to get Wade, an 8-year-old autistic boy, to make eye contact when she speaks to him. She gives Wade a piece of candy every time he looks at her face. This treatment illustrates which of the following therapeutic approaches?
- (A) Cognitive
 - (B) Biological
 - (C) Psychodynamic
 - (D) Humanistic
 - (E) Behavioral
19. Which of the following forms of therapy most likely involves a confrontational atmosphere between the therapist and the client?
- (A) Rational-emotive therapy
 - (B) Psychoanalysis
 - (C) Aversive conditioning
 - (D) Person-centered therapy
 - (E) Systematic desensitization
20. A person who has a brain injury is having difficulty seeing and hearing. These symptoms indicate that damage has occurred in the
- (A) parietal and occipital lobes
 - (B) occipital and temporal lobes
 - (C) frontal and temporal lobes
 - (D) temporal lobe only
 - (E) frontal lobe only
21. Individuals who accept personal responsibility for their life experiences may be characterized as having
- (A) unrealistic expectations
 - (B) delusions of grandeur
 - (C) an internal locus of control
 - (D) a pessimistic view of reality
 - (E) an introverted personality
22. A student participates in a month-long sleep study designed to examine free-running circadian rhythms. If all time cues are removed, the student's total sleep-wake cycle is likely to
- (A) average about 25 hours
 - (B) average about 12 hours
 - (C) average whatever it had averaged when the student began the study
 - (D) become even more dependent than usual on the student's activity level
 - (E) become extremely variable
23. If the variance of a set of scores is 100, the standard deviation will be
- (A) 5
 - (B) 10
 - (C) 25
 - (D) 50
 - (E) 125
24. The overall purpose of psychoanalytic therapy is to
- (A) change the behavior patterns of the client in therapy
 - (B) help the client in therapy reach his or her full potential
 - (C) alter the thought processes of the client in therapy
 - (D) help the client in therapy gain new insight into himself or herself
 - (E) help the client in therapy get rid of irrational thoughts

25. For a language test with normally distributed scores, the mean was 70 and the standard deviation was 10. Approximately what percentage of test takers scored 60 and above?
- (A) 16
 - (B) 34
 - (C) 68
 - (D) 84
 - (E) 95
26. Which of the following are most involved in the action potential of a neuron?
- (A) Calcium and sodium
 - (B) Sodium and potassium
 - (C) Potassium and calcium
 - (D) Chloride and calcium
 - (E) Chloride and sodium
27. Which of the following would be used to measure the relationship between age and reaction time?
- (A) Correlation
 - (B) Central tendency
 - (C) A histogram
 - (D) Standard deviation
 - (E) A t test
28. Which component of the limbic system has an essential role in the formation of new memories?
- (A) Amygdala
 - (B) Hippocampus
 - (C) Pituitary gland
 - (D) Hypothalamus
 - (E) Thalamus
29. The human vestibular sense is most closely associated with the
- (A) skin
 - (B) semicircular canals
 - (C) taste buds
 - (D) olfactory bulb
 - (E) rods and cones
30. A central nervous system depressant that produces a false feeling of well-being and efficiency and results in slower reaction time to stimulation is
- (A) cocaine
 - (B) marijuana
 - (C) dopamine
 - (D) alcohol
 - (E) nicotine
31. Of the following research methods, which can best establish a cause-and-effect relationship?
- (A) Naturalistic observation
 - (B) A survey
 - (C) A test
 - (D) A case study
 - (E) An experiment
32. A researcher dabs color on a 16-month-old child's face and places the child in front of a mirror. Which of the following developmental milestones has been reached if the child realizes that there is something wrong with its face?
- (A) Visual discrimination
 - (B) Recognition of a human form
 - (C) Recognition of self
 - (D) Identification of the gender of the image
 - (E) Perception of the image as a playmate
33. A monkey is conditioned to flinch at the sound of a bell that was previously paired with a puff of air to the monkey's cheek. Which of the following explanations would be consistent with a cognitive interpretation of this conditioning?
- (A) The animal cannot control its tendency to flinch because the response of flinching is simply a reflex to the bell.
 - (B) The strength of the flinch response is a function of the time interval between the onset of the bell and the air puff.
 - (C) The monkey interprets the bell as a signal that the air puff will follow.
 - (D) The bell is merely a substitute stimulus for the air puff.
 - (E) Monkeys are intelligent and know that they should flinch when they hear tones that are paired with stimuli that elicit reflexes.

34. Alfred Binet's most important contribution to psychology was in the area of
- (A) intelligence testing
 - (B) visual perception
 - (C) psychopathology
 - (D) comparative psychology
 - (E) classical conditioning
35. Which of the following is most likely to characterize the behavior of students who have high achievement motivation and are intrinsically motivated to play a musical instrument?
- (A) If promised a reward for practicing a difficult piece of music selected by the teacher, they will practice more than if they selected the piece by themselves.
 - (B) If permitted to choose their own pieces of music, they will select very difficult ones that are beyond their present ability to play.
 - (C) If permitted to choose their own pieces of music, they will select easy pieces that they can master in one practice session.
 - (D) If permitted to choose their own pieces of music, they will select moderately difficult pieces that they can master if they practice them conscientiously.
 - (E) They will enjoy practicing the piano more if their parents promise them a reward for mastering each piano piece.
36. Which of the following is considered an explanation of why bystander intervention is less likely to occur if there is a large number of witnesses to a crime?
- (A) Prejudice
 - (B) Social facilitation
 - (C) Diffusion of responsibility
 - (D) Group polarization
 - (E) Self-efficacy
37. A student who strongly believes that genetic influence is the major contributor to human personality is analyzing data gathered about identical twins who had been separated at birth and reunited in adulthood. The student observes many striking similarities in personality and habits within the twin pairs but does not notice differences within the twin pairs that might argue against the student's belief.
- This student's behavior illustrates which of the following?
- (A) Confirmation bias
 - (B) The availability heuristic
 - (C) An algorithmic error
 - (D) Metacognition
 - (E) A mnemonic
38. A schema can be described as
- (A) an outer layer of the eye
 - (B) a mental construct
 - (C) a fissure between lobes of the brain
 - (D) an optical illusion
 - (E) a fixed response to a particular stimulus
39. John Garcia showed that when rats ingested a novel substance before becoming nauseated from radiation or drugs, they acquired a
- (A) conditioned taste preference for the substance
 - (B) generalized taste preference for similar substances
 - (C) conditioned taste aversion for the substance
 - (D) conditioned taste aversion for any novel substance
 - (E) conditioned taste preference for any novel substance

Section I

40. Which of the following is a possible reason why cats can see better at night than can humans?
- (A) Cats have a higher proportion of rods to cones.
 - (B) Cats' pupils can contract to a smaller opening.
 - (C) Cats have a smaller blind spot.
 - (D) Cats have a larger optic nerve tract.
 - (E) The visual cortex of cats is located farther forward in the cortex.
41. Which of the following reflects the inborn and stable rudiments of personality, such as excitability?
- (A) Imprinting
 - (B) Social referencing
 - (C) Individuation
 - (D) Temperament
 - (E) Accommodation
42. After seeing her parents give her brother a dollar for cleaning his room, Sarah begins cleaning her own room. According to social-learning theorists, Sarah's behavior is an example of which of the following?
- (A) Classical conditioning
 - (B) Spontaneous recovery
 - (C) Stimulus generalization
 - (D) Discrimination training
 - (E) Observational learning
43. In Abraham Maslow's hierarchy of needs, living up to one's potential and striving for personal fulfillment are referred to as
- (A) biological needs
 - (B) aesthetic needs
 - (C) physiological needs
 - (D) belonging needs
 - (E) self-actualization needs
44. The technique of assessing personality by asking a person to make up a story based on a picture presented by the researcher is an example of which of the following types of tests?
- (A) Trait/type
 - (B) Projective
 - (C) Forced choice
 - (D) Simulation
 - (E) Personality inventory
45. John B. Watson was a pioneer in which of the following perspectives of psychology?
- (A) Biological
 - (B) Functionalism
 - (C) Psychoanalytic
 - (D) Structuralism
 - (E) Behaviorism
46. Which of the following is considered the fundamental building block of the nervous system?
- (A) Nucleus
 - (B) Neuron
 - (C) Synapse
 - (D) Neurotransmitter
 - (E) Electrical impulse
47. The psychological effects of alcohol are powerfully influenced by the user's
- (A) expectations
 - (B) success in developing a social network
 - (C) agility
 - (D) intelligence quotient (IQ)
 - (E) brain dopamine level

48. Solomon Asch's findings on conformity might best be used to explain why
- (A) members of a family all like the taste of bananas
 - (B) adolescents follow fads in dress and hairstyle
 - (C) people are less likely to accept blame for their failures than to accept credit for their successes
 - (D) bystander intervention is more likely to occur when few, rather than many, bystanders are present
 - (E) performance is enhanced in the presence of others
49. The humanistic perspective in psychology suggests that
- (A) most behavior can be explained by operant conditioning principles
 - (B) all humans have peak experiences
 - (C) all humans are by nature good
 - (D) humans are doomed to a life of neurosis and suffering
 - (E) self-actualization can happen only through therapy
50. The human brain differs from the brains of most other animals by the relative amount of brain mass devoted to which of the following?
- (A) The occipital lobe
 - (B) The cerebellum
 - (C) The cerebral cortex
 - (D) The homunculus
 - (E) The pituitary gland
51. Which of the following is an example of retrograde amnesia?
- (A) Ty cannot recall the face of the thief he saw running from the scene of the crime.
 - (B) Cassie's vivid memory of the explosion of the space shuttle Challenger is not corroborated by those she was with at the time.
 - (C) Alberto is unable to remember anything since the accident that destroyed portions of his hippocampus.
 - (D) Katie attributes her poor performance on a standardized test to the fact that she took the exam in a room other than the one in which she learned the material.
 - (E) Alyse cannot remember any details of what happened right before her car accident.
52. An individual who sees and feels imaginary spiders crawling on his arms and legs is experiencing
- (A) a fixation
 - (B) a hallucination
 - (C) an illusion
 - (D) an eidetic image
 - (E) a phobia
53. Research indicates that many animals are more likely to associate sickness with a taste they experienced in conjunction with the illness than with a tone or light. This finding supports which of the following claims?
- (A) The tone or light must not have been appropriately paired with the onset of illness.
 - (B) Illness is not necessarily punishing to subjects.
 - (C) Animals may be biologically prepared to learn some things over other things.
 - (D) Extrinsic reinforcers may be more effective than intrinsic reinforcers.
 - (E) Positive reinforcers are more effective than punishers.

Section I

54. Which of the following is associated with schizophrenia?
- (A) Enlarged, fluid-filled areas in the brain
 - (B) Damage to the medulla
 - (C) Malfunction of the endocrine system
 - (D) Impairment of the spinal reflexes
 - (E) Injury to the parasympathetic nervous system
55. According to the psychoanalytic view, depression is caused by
- (A) a neurotransmitter imbalance
 - (B) self-defeating thoughts
 - (C) prolonged exposure to stressors over which the individual has limited control
 - (D) unresolved experiences of loss from childhood
 - (E) alcohol or other substance abuse
56. Which of the following structures of the brain has been linked with the regulation of hunger and thirst?
- (A) Hippocampus
 - (B) Hypothalamus
 - (C) Thalamus
 - (D) Pons
 - (E) Medulla
57. Sigmund Freud once compared psychological development to the migration of a population: "As time passes most of the group moves on, but some of the members remain at an earlier location."
- Which concept was Freud describing?
- (A) Fixation
 - (B) Repression
 - (C) Regression
 - (D) Sublimation
 - (E) Reaction formation
58. One perspective in clinical psychology proposes that adaptive and abnormal behaviors can be developed through similar processes. Which of the following terms best characterizes this approach to abnormal behavior?
- (A) Biological
 - (B) Psychodynamic
 - (C) Behavioral
 - (D) Humanistic
 - (E) Cognitive
59. Which of the following treatments is most frequently used to eliminate specific phobias?
- (A) Antidepressant drugs
 - (B) Systematic desensitization
 - (C) Implosion therapy
 - (D) Psychoanalysis
 - (E) Aversion therapy
60. Which of the following terms is used in hunger and weight-control research to denote the concept that each person has a body-fat level that remains fixed and resistant to change?
- (A) Hyperphagia
 - (B) Hypophagia
 - (C) Glucagon theory
 - (D) Set point
 - (E) Metabolic conversion
61. A 14-month-old toddler is placed in an unfamiliar situation with the child's mother, who then leaves the room for a time. When the mother returns, the child squirms and tries to get away from the mother when picked up, but also seems distressed when placed back on the floor. Mary Ainsworth would consider this evidence of which of the following?
- (A) Hyperactivity
 - (B) Narcissistic personality type
 - (C) A resistant or ambivalent attachment style
 - (D) Disorganized behavior
 - (E) Avoidance

62. The two components of Schachter and Singer's theory of emotion are
- (A) love and anger
 - (B) instinct and motivation
 - (C) instinct and brain activation
 - (D) physiological arousal and cognitive assessment
 - (E) physiological arousal and behavior
63. Which of the following sets of scores has the greatest standard deviation?
- (A) 5, 7, 9, 12
 - (B) 2, 7, 9, 22
 - (C) 25, 27, 29, 32
 - (D) 50, 51, 52, 53
 - (E) 100, 101, 101, 102, 103
64. A college student experiences a loss of sensation in her right arm before exams. Doctors can find no physiological basis for her condition. This student is most likely experiencing which of the following kinds of disorders?
- (A) Somatoform
 - (B) Dissociative
 - (C) Anxiety
 - (D) Mood
 - (E) Personality
65. Rats in an experiment learned to associate sweetened water with a drug that causes immune suppression. Later, the sweetened water alone produced the immune suppression. This outcome is an example of which of the following?
- (A) Learned helplessness
 - (B) Systematic desensitization
 - (C) Operant conditioning
 - (D) Classical conditioning
 - (E) Biofeedback
66. Early behaviorists believed that psychology should NOT focus on "the mind" because "the mind" is
- (A) too complex
 - (B) genetically determined
 - (C) largely unconscious
 - (D) unobservable
 - (E) environmentally determined
67. Which of the following explains the behavior of normally law-abiding people who act destructively when they are part of a crowd?
- (A) Group polarization
 - (B) The mere exposure effect
 - (C) Deindividuation
 - (D) Entrapment
 - (E) Fundamental attribution error
68. The psychologists who first developed encounter groups and sensitivity-training groups based their work on which of the following approaches to therapy?
- (A) Psychodynamic
 - (B) Humanistic
 - (C) Behavioral
 - (D) Cognitive
 - (E) Biological
69. Greg stays up all night during finals week studying for exams. As the week progresses, his muscles tighten and he develops a stiff neck. By the last day of finals, he is taking more frequent breaks, leaning back in the desk chair, and staring off into space. He arrives for the last test with a sore throat and headache.
- Which of the following best describes Greg's response to stress?
- (A) General adaptation syndrome
 - (B) Object-relations theory
 - (C) Opponent-process theory
 - (D) Two-factor theory
 - (E) Type B behavior pattern

70. Which of the following is evidence of the reliability of a new intelligence test?
- (A) A correlation of $+0.90$ exists between scores on the new test and scores on the Wechsler Intelligence Scale for Children.
 - (B) The test predicts students' ability to succeed in college.
 - (C) The correlation between scores for identical twins taking the test is $+0.90$.
 - (D) Baseline data for test norming are obtained from a diverse sample of several thousand participants.
 - (E) The correlation between scores of participants who take two forms of the test is $+0.90$.
71. The practice of solving problems by using a mental shortcut is an example of
- (A) an insightful operation
 - (B) a confirmation bias
 - (C) a hypothesis test
 - (D) the use of a heuristic
 - (E) the use of an algorithm
72. The minimum intensity at which a stimulus can be detected at least 50 percent of the time is known as the
- (A) visual cliff
 - (B) just noticeable difference
 - (C) perceptual set
 - (D) receptor potential
 - (E) absolute threshold
73. Remembering how to roller skate involves which of the following kinds of memory?
- (A) Semantic
 - (B) Episodic
 - (C) Priming
 - (D) Procedural
 - (E) Prospective
74. Prozac functions as an antidepressant medication because it
- (A) enhances production of acetylcholine
 - (B) blocks the reuptake of the neurotransmitter serotonin
 - (C) causes selective memory loss for depression-producing events
 - (D) produces a steady, mild state of euphoria
 - (E) inhibits frontal lobe activity related to depression
75. A club president discovers that contributions of club members drop when the total contribution of all members is published rather than the contributions of individuals. This drop can be explained by the phenomenon of
- (A) group polarization
 - (B) learned helplessness
 - (C) social loafing
 - (D) social facilitation
 - (E) socialization
76. A researcher studies the effects of brain lesions in rats. This research study reflects which of the following perspectives?
- (A) Cognitive
 - (B) Biological
 - (C) Behavioral
 - (D) Humanistic
 - (E) Psychoanalytic
77. Material that an individual cannot remember but is on the "tip of the tongue" is
- (A) in episodic memory, but not in semantic memory
 - (B) in sensory memory, but not in iconic memory
 - (C) in short-term memory, but not in long-term memory
 - (D) available, but not accessible
 - (E) retrieved, but not encoded

78. A teacher taught her students to take turns by giving them stars to trade for snacks at the end of the day. This technique is called
- (A) systematic desensitization
 - (B) token economy
 - (C) classical conditioning
 - (D) rational-emotive therapy
 - (E) free association
79. Which Piagetian stage of cognitive development is characterized by mastery of conservation tasks?
- (A) Sensorimotor
 - (B) Preoperational
 - (C) Concrete operations
 - (D) Formal operations
 - (E) Tertiary circular reactions
80. Which of the following do individuals experience when their behavior is inconsistent with their attitude?
- (A) Approach-avoidance conflict
 - (B) Cognitive dissonance
 - (C) Intrinsic motivation
 - (D) Homeostatic motivation
 - (E) Overjustification
81. The evolutionary perspective (sociobiology) argues that
- (A) humans are motivated by a desire to maintain optimum hormonal levels
 - (B) humans are genetically programmed to ensure that there is a new generation of the species
 - (C) humans strive to minimize inconsistent thoughts and actions
 - (D) there is a relationship among task difficulty, physical arousal, and task performance
 - (E) conditioning explains most human behavior
82. When Jason practices the drums, he tends not to hear the phone. Today he is expecting a call from a record producer and answers the phone each time it rings even when he is practicing the drums. Which of the following explains why Jason hears the phone today?
- (A) Weber's law
 - (B) Accommodation
 - (C) Frequency theory
 - (D) Signal detection theory
 - (E) Harmonics
83. Which of the following is NOT a Gestalt principle of perceptual organization?
- (A) Proximity
 - (B) Similarity
 - (C) Closure
 - (D) Intensity
 - (E) Continuity
84. Which of the following is the correct sequence of anatomical structures through which an auditory stimulus passes before it is perceived as sound?
- (A) Cochlea, ossicles, eardrum, oval window, auditory canal
 - (B) Eardrum, cochlea, auditory canal, ossicles, oval window
 - (C) Oval window, auditory canal, eardrum, cochlea, ossicles
 - (D) Ossicles, eardrum, cochlea, auditory canal, oval window
 - (E) Auditory canal, eardrum, ossicles, oval window, cochlea

Section I

85. Aaron Beck suggested that negative beliefs cause depression. To help change these negative beliefs, Beck used which of the following therapies?
- (A) Cognitive
 - (B) Psychopharmacological
 - (C) Rational-emotive
 - (D) Psychoanalytic
 - (E) Social-learning
86. When Shelly first had cable television service installed, Public Broadcasting (PBS) was on channel 9. Her cable company then switched PBS to channel 16. Shelly now has trouble remembering that PBS is on channel 16 and not on channel 9. This memory problem represents
- (A) memory decay
 - (B) retrograde amnesia
 - (C) reconstruction errors
 - (D) retroactive interference
 - (E) proactive interference
87. Two friends attribute their high math scores to their high level of effort and ability in math and their low Spanish scores to their teacher's subjective grading and favoritism. In this situation these students are exhibiting
- (A) the actor-observer effect
 - (B) the hindsight bias
 - (C) scapegoating
 - (D) equity
 - (E) the self-serving bias
88. All of the following are conditions that may lead to conflict within organizations EXCEPT
- (A) scarce resources
 - (B) jurisdictional ambiguity
 - (C) inequities in status
 - (D) insufficient communication
 - (E) superordinate goals
89. Which of the following is most descriptive of antisocial personality disorder?
- (A) A pattern of limited social interaction and reluctance to enter into relationships
 - (B) A pattern of extreme dependence on other people and acute anxiety at being left alone
 - (C) A pattern of bizarre or unstable behavior characterized by dramatic mood shifts
 - (D) An inability to feel empathy for others and a lack of remorse for actions that harm others
 - (E) An exaggerated sense of self-importance
90. Which of the following assessment tools is LEAST likely to be used by a psychoanalytically oriented psychologist?
- (A) Sentence completion
 - (B) Minnesota Multiphasic Personality Inventory
 - (C) Draw-A-Person Test
 - (D) Thematic Apperception Test
 - (E) Rorschach Test
91. Which of the following neurotransmitters is most directly associated with Alzheimer's disease?
- (A) Dopamine
 - (B) Serotonin
 - (C) Gamma-aminobutyric acid (GABA)
 - (D) Acetylcholine (ACh)
 - (E) Glutamate

Questions 92-93 refer to the following psychological perspectives. Choose the perspective that best answers each of the following questions. A choice can be used once, more than once, or not at all.

- (A) Humanistic
- (B) Trait
- (C) Psychoanalytic
- (D) Behavioral
- (E) Cognitive

92. Which perspective views the human condition as a mixture of unconscious desires and conflicts?

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

93. Which perspective was used by Raymond Cattell as a basis for his personality measure known as the 16PF test?

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

94. According to the theory of fundamental attribution error, when explaining the failures of others we usually underestimate the significance of

- (A) situational factors
- (B) dispositional factors
- (C) motivational factors
- (D) support systems
- (E) inherited traits

95. Language acquisition cannot be fully accounted for by associative learning processes for which of the following reasons?

- (A) Language use is creative.
- (B) Speakers construct rules for utterances by imitating the models they hear.
- (C) Effective communication depends on one's level of emotional experience.
- (D) Language production is reinforced by the listener.
- (E) Infants are too young to learn associatively.

Section I

Questions 96-98 refer to the following study.

A student hypothesizes that high school students consuming different flavors of a drink before a spelling test will perform differently. A study to test the hypothesis finds that with a bitter drink, performance is best 6 hours after drinking it, whereas with a sweet drink, performance is better 1 hour after drinking it.

96. Which of the following are the independent variables?

- (A) Test scores and high school students
- (B) Test scores and time of consumption
- (C) Flavor of drink and time of consumption
- (D) Flavor of drink and high school students
- (E) Flavor of drink and test scores

97. Which of the following is the dependent variable?

- (A) Flavor of drink
- (B) Participants' spelling test scores
- (C) Participants' ages
- (D) Time the drink was consumed
- (E) Number of drinks consumed

98. An interaction between variables complicates the researcher's explanation of findings. Which of the following are most likely involved in this interaction?

- (A) Test scores and high school students
- (B) Test scores and time of consumption
- (C) Flavor of drink and time of consumption
- (D) Flavor of drink and high school students
- (E) Flavor of drink and test scores

99. Which of the following is characterized by a periodic appearance of sleep spindles?

- (A) Stage 2 sleep
- (B) Stage 3 sleep
- (C) Stage 4 sleep
- (D) REM sleep
- (E) Night terrors

100. Social facilitation theory focuses on situations in which the presence of others causes an individual's performance to

- (A) remain unchanged
- (B) decline
- (C) improve
- (D) spontaneously recover
- (E) become fixated

END OF SECTION I

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PSYCHOLOGY

SECTION II

Time—50 minutes

Percent of total grade— $33\frac{1}{3}$

Directions: You have 50 minutes to answer BOTH of the following questions. It is not enough to answer a question by merely listing facts. You should present a cogent argument based on your critical analysis of the questions posed, using appropriate psychological terminology.

1. Dr. Franklin investigated the relationship between stress and physical illness. She persuaded a high school principal to require all female athletes in the school to participate in her study. She explained the purpose of her research to the athletes and asked them to use a standard form to rate the severity of their stress over the last six months. More than 250 athletes completed the form. Then Dr. Franklin analyzed the forms returned by the first 100 athletes. She requested the attendance records from the nurse's office for each of those athletes to verify the number of days absent due to illness during the same six-month period. In the debriefing summary that Dr. Franklin sent to the 100 athletes after she completed the study, she stated that athletes who reported more stress also experienced more frequent illness. Dr. Franklin concluded that stress causes physical illness.
 - Identify the research method used by Dr. Franklin.
 - Describe the operational definitions of the TWO key variables that Dr. Franklin used in the study.
 - Based on the information provided, describe ONE appropriate and ONE inappropriate ethical feature of the study.
 - Identify ONE statistical technique that Dr. Franklin could use to represent the relationship between the variables in the study.
 - Describe TWO aspects of research design that weaken the validity of Dr. Franklin's conclusion that stress causes physical illness.

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2. Time is an important variable in many psychological concepts. Describe a specific example that clearly demonstrates an understanding of each of the following concepts and how it relates to or is affected by time. Use a different example for each concept.
- Critical period
 - Fluid intelligence
 - Group polarization
 - James-Lange theory of emotion
 - Presentation of the conditioned stimulus (CS) and unconditioned stimulus (UCS) in classical conditioning
 - Refractory period in neural firing
 - Sound localization
 - Spontaneous recovery

END OF EXAMINATION

Chapter III: Answers to the 2004 AP Psychology Exam

■ Section I: Multiple Choice

- Section I Answer Key and Percent Answering Correctly
- Analyzing Your Students' Performance on the Multiple-Choice Section
- Diagnostic Guide for the 2004 AP Psychology Exam

■ Section II: Free Response

- Comments from the Chief Reader
- Scoring Guidelines, Sample Student Responses, and Commentary
 - Question 1
 - Question 2

Section I: Multiple Choice

Listed below are the correct answers to the multiple-choice questions, the percent of AP students who answered each question correctly by AP grade, and the total percent answering correctly.

Section I Answer Key and Percent Answering Correctly

Item No.	Correct Answer	Percent Correct by Grade					Total Percent Correct
		5	4	3	2	1	
1	B	98	97	95	94	90	95
2	D	100	99	98	96	88	97
3	A	97	92	83	71	48	82
4	E	97	92	87	79	62	86
5	D	85	76	67	55	40	68
6	A	99	97	94	88	78	93
7	C	99	97	93	85	62	91
8	C	93	85	76	62	42	76
9	C	99	96	90	80	64	89
10	D	86	80	74	67	52	74
11	E	94	85	75	64	46	77
12	C	97	93	88	80	60	87
13	D	89	84	77	67	50	76
14	B	92	87	80	70	50	79
15	D	93	84	75	67	53	78
16	A	88	77	66	55	40	69
17	C	96	91	86	81	70	87
18	E	98	94	87	76	56	86
19	A	53	36	24	17	13	31
20	B	93	84	77	66	50	77
21	C	98	91	81	65	42	80
22	A	84	67	51	34	17	56
23	B	40	32	28	26	24	31
24	D	90	79	65	52	37	69
25	D	46	29	22	18	14	27

Item No.	Correct Answer	Percent Correct by Grade					Total Percent Correct
		5	4	3	2	1	
26	B	71	58	47	38	28	51
27	A	89	84	80	75	59	80
28	B	89	77	65	52	34	68
29	B	74	50	34	21	15	43
30	D	81	69	61	53	38	64
31	E	98	92	87	78	65	87
32	C	85	76	70	67	59	73
33	C	88	81	76	72	58	77
34	A	96	89	78	64	41	78
35	D	74	63	57	52	42	60
36	C	96	88	77	62	36	77
37	A	99	96	92	82	60	89
38	B	98	94	83	63	32	80
39	C	85	72	59	46	31	63
40	A	92	83	74	64	47	76
41	D	89	72	55	39	25	61
42	E	98	96	93	88	76	92
43	E	99	97	94	88	70	92
44	B	90	78	67	57	45	71
45	E	79	68	58	46	32	60
46	B	95	87	79	66	43	78
47	A	78	60	47	34	20	52
48	B	84	75	68	59	37	68
49	C	93	84	73	58	33	73
50	C	86	70	57	46	35	62

(continued)

Section I Answer Key and Percent Answering Correctly (continued)

Item No.	Correct Answer	Percent Correct by Grade					Total Percent Correct
		5	4	3	2	1	
51	E	72	64	58	51	38	59
52	B	98	95	93	89	78	92
53	C	85	63	43	26	18	52
54	A	71	55	45	36	26	49
55	D	97	89	73	49	21	72
56	B	90	79	70	59	44	72
57	A	78	66	54	42	28	57
58	C	72	61	54	48	38	57
59	B	97	92	83	67	38	80
60	D	93	85	75	64	42	76
61	C	88	78	69	60	48	72
62	D	79	52	33	21	16	44
63	B	92	84	76	65	47	76
64	A	81	66	48	33	19	54
65	D	81	69	57	45	28	60
66	D	88	76	64	48	27	65
67	C	85	68	51	35	21	56
68	B	55	46	39	31	24	41
69	A	66	40	27	19	17	37
70	E	74	55	42	32	20	48
71	D	89	75	62	46	27	64
72	E	89	79	70	59	39	71
73	D	85	72	62	51	37	65
74	B	85	72	61	49	36	64
75	C	87	69	55	43	32	61

Item No.	Correct Answer	Percent Correct by Grade					Total Percent Correct
		5	4	3	2	1	
76	B	95	88	80	68	47	80
77	D	78	62	50	38	26	55
78	B	96	88	76	60	41	77
79	C	76	63	51	40	28	55
80	B	97	90	77	60	36	77
81	B	91	79	65	50	29	67
82	D	70	54	47	41	35	51
83	D	85	68	54	40	26	59
84	E	69	54	46	38	28	50
85	A	57	47	37	26	16	40
86	E	61	40	29	21	15	36
87	E	88	79	72	61	42	72
88	E	73	56	44	37	27	50
89	D	87	69	50	32	18	56
90	B	83	58	39	23	15	48
91	D	60	49	42	36	26	45
92	C	99	97	87	68	36	83
93	B	73	53	42	34	27	49
94	A	94	83	71	58	36	73
95	A	31	17	11	8	7	16
96	C	87	74	61	45	29	64
97	B	97	90	79	63	38	78
98	C	46	34	28	24	21	32
99	A	53	37	27	20	12	32
100	C	91	79	69	56	38	71

Analyzing Your Students' Performance on the Multiple-Choice Section

If you give your students the 2004 AP Psychology Exam for practice, you may want to analyze the results to find overall strengths and weaknesses in their understanding of AP Psychology. The following diagnostic worksheet will help you do this. You are permitted to photocopy and distribute it to your students for completion.

1. In each section, students should insert a check mark for each correct answer.
2. Add together the total number of correct answers for each section.

3. To compare the student's number of correct answers for each section with the average number correct for that section, copy the number of correct answers to the "Number Correct" table at the end of the Diagnostic Guide.

In addition, under each item, the percent of AP students who answered correctly is shown, so students can analyze their performance on individual items. This information will be helpful in deciding how students should plan their study time.

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Question 1

Dr. Franklin investigated the relationship between stress and physical illness. She persuaded a high school principal to require all female athletes in the school to participate in her study. She explained the purpose of her research to the athletes and asked them to use a standard form to rate the severity of their stress over the last six months. More than 250 athletes completed the form. Then Dr. Franklin analyzed the forms returned by the first 100 athletes. She requested the attendance records from the nurse's office for each of those athletes to verify the number of days absent due to illness during the same six-month period. In the debriefing summary that Dr. Franklin sent to the 100 athletes after she completed the study, she stated that athletes who reported more stress also experienced more frequent illness. Dr. Franklin concluded that stress causes physical illness.

- Identify the research method used by Dr. Franklin.
- Describe the operational definitions of the TWO key variables that Dr. Franklin used in the study.
- Based on the information provided, describe ONE appropriate and ONE inappropriate ethical feature of the study.
- Identify ONE statistical technique that Dr. Franklin could use to represent the relationship between the variables in the study.
- Describe TWO aspects of research design that weaken the validity of Dr. Franklin's conclusion that stress causes physical illness.

General Issues

1. Acceptable answers must include sentences (subject/verb): no stand-alone pictures or outlines.
2. You may establish which point is being discussed by the structure of the essay if it is written in order of the questions asked.
3. A point once earned cannot be lost by a later wrong statement unless immediately contradicted.

POINT 1: Research Method

- correlation
- survey
- questionnaire

NOTES: Ignore use of the word "experiment" as a synonym for study. "In this experiment, Dr. Franklin used a survey" counts. However, the specific phrase "correlational experiment" is a direct contradiction and will not be counted.

Don't Score:

- "experiment" as the method ("Dr. Franklin conducted an experiment that used surveys.")
- longitudinal study
- case study
- observational study

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Question 1 (cont'd.)

POINT 2: Operational Definition of Stress

- stress severity rating
- stress questionnaire
- stress score
- self-report of stress ("self-report" implies a survey was used)

NOTES: a) The operational definition requires identification of the method by which stress was measured (i.e., reference to the stress instrument used by Franklin).

b) A good operational definition of stress can use the exact language from the question.

Don't Score:

- "stress," "stress level," or "amount of stress"

POINT 3: Operational Definition of Illness

- number of days absent (don't need to specify "due to illness" or "six-month period")
- measured from attendance records

NOTE: A good operational definition of illness can use the exact language from the question.

Don't Score:

- "number of days sick" (illness wasn't directly measured, attendance was)
- procedures (e.g., "asked when they were sick")
- definition of illness (can't just describe illness or list symptoms)

POINT 4: Appropriate Ethical Characteristic (*must be based on information provided*)

- shared statement of purpose with participants
- debriefed participants

Don't Score:

- ethical practices not mentioned in question (e.g., providing confidentiality, getting IRB approval)

POINT 5: Inappropriate Ethical Characteristic (*must be based on information provided*)

- debriefed only some (100 of 250) participants (can't get both points 4 & 5 for saying "only some debriefed")
- were forced to participate (required by principal) / weren't allowed to quit
- no consent (student or parent)
- didn't get participant's permission to look at attendance records (invasion of privacy)

Don't score:

- poor design characteristics (e.g., "used only 100 when she could have used 250")
- did not use an IRB procedure (does not restrict answer to details provided in the question)

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Question 1 (cont'd.)

POINT 6: Statistical Technique

- calculate the correlation
- correlation coefficient
- scatter plot / correlation graph (“graphing stress on one axis and absences on the other”)
- calculate regression line

Don't score:

- analysis of variance, t test, z score, chi square test, inferential statistics

POINTS 7 and 8: Two Design Features that Weaken the Conclusion

Two parts required for each point: Identifying the problem and saying how or why it is a weakness. The two features cannot be the same issue (both points can't come from same bullet below).

- **causation** is not established
 - correlation is not causation (“why” implied) NOTE: can be imbedded in point 6, but not in others
 - direction of cause could be reversed (“illness could cause stress”)
 - 3rd variable: Illness could be caused by a variable other than stress
 - “illness could be caused by another variable that Franklin didn't measure”
 - example of a 3rd variable that affects the health of all participants, not only particular individuals: “absences could have been caused by the fear of violence in the school”
 - Don't score: “a student who plays soccer has more stress than others”
- testing only girls (or athletes, high school students) limits ability to **generalize** or is not **representative**
- took only the first 100 respondents (sample problem), so didn't have a **random sample**
- should have used more precise measures of stress or illness to improve the **operational definition** (“the illness measure is bad because absences may be due to things other than illness”)
- **response bias** (specified problem tied to “why” of response bias)
 - “Franklin told participants about the research before she collected data, biasing the results.”
 - “Students might lie about their stress, producing inaccurate data.”
 - Surveys, in general, are open to social desirability (trying to look good), biasing responses.

Don't Score:

- “no random assignment”
- “boys weren't allowed to participate” without link to generalization or representativeness
- “sample size too small”
- can't just disagree with Dr. Franklin's conclusion--must identify design problem (e.g., “can't conclude that stress causes physical illness because it is not an experiment” scores only with underlined portion)

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Question 2

Time is an important variable in many psychological concepts. Describe a specific example that clearly demonstrates an understanding of each of the following concepts and how it relates to or is affected by time. Use a different example for each concept.

- Critical period
- Fluid intelligence
- Group polarization
- James-Lange theory of emotion
- Presentation of the conditioned stimulus (CS) and unconditioned stimulus (UCS) in classical conditioning
- Refractory period in neural firing
- Sound localization
- Spontaneous recovery

General Considerations

1. The two overriding criteria for scoring each point are to demonstrate understanding of the concept and to relate the concept to time.
2. Because it is difficult to demonstrate understanding of some concepts (for example, the refractory period) with an example, well-phrased descriptions or definitions may be substituted for examples. However:
 - a. if a description and an example *directly contradict* one another, do not score the point.
 - b. if a good example is coupled with a weak or ambiguous description (or vice versa), score the point.
3. Information must be presented in context. That is, it must be clear that the student is discussing a particular point before the point can be awarded.
4. Answers must be presented in sentences, and sentences must be cogent enough for student's meaning to come through. Spelling and grammatical mistakes do not reduce a student's score, but spelling must be close enough so that the reader is convinced of the word.
5. A student will not be penalized for misinformation unless it directly contradicts correct information that would otherwise have scored a point.

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Question 2 (cont'd.)

Point 1: Critical period

The student **must both**:

1. establish the vital nature of critical periods in the developmental process by a description or example of how:
 - a. the selected critical period is *necessary* for healthy development, or
 - b. a disruption of the normal developmental process during the critical period leads to abnormal development, or
 - c. greater sensitivity to outside influences (such as drugs) during the critical period can negatively impact development.
2. identify an appropriate developmental time period to establish the critical period. For example, the time period may be identified as “childhood.”

Typical examples:

1. “Baby ducks must imprint and begin to follow their mothers during a critical period shortly after their birth.”
2. “Children are predisposed to learn language easily during childhood.”
3. “Exposure to teratogens like drugs can interfere with normal prenatal development.”

Traps:

1. Do not score the point if the student describes a time period but is missing the critical nature of the development which is to occur during that window (“Children learn language during childhood” does not score).
2. Do not score the point if a concept or theory (e.g., “imprinting,” “language acquisition theory”) is merely mentioned by name without a description or illustrating example.

Point 2: Fluid intelligence

The student **must both**:

1. include a *dynamic cognitive component* (e.g., reasoning ability, problem solving, flexibility, speedy information processing, etc.) in the example or description of fluid intelligence.
2. establish a time relationship appropriate to fluid intelligence by describing:
 - a. the decline of fluid intelligence over time, or
 - b. how people with greater fluid intelligence require less time to complete cognitive tasks such as problem solving.

Typical examples:

1. “Fluid intelligence, the ability to think and solve problems, decreases as one ages.”
2. “People with less fluid intelligence take more time to solve puzzles than people with more fluid intelligence.”

Trap: Do not score the point if the student confuses fluid intelligence with crystallized intelligence or general knowledge.

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Question 2 (cont'd.)

Point 3: Group polarization

The student **must** both:

1. establish that group polarization means that a group's predominant view, attitude, or opinion becomes stronger or more extreme. It is important that the student get across the idea of an intensification of a group's opinion rather than a person changing his or her attitude to become more like the group (conformity).
2. demonstrate an understanding that this happens after or as a result of time for group interaction or communication.

Typical examples:

1. "A jury often becomes more convinced of a defendant's guilt or innocence as deliberations continue."
2. "Prejudicial attitudes become more extreme when prejudiced people talk with each other."
3. "Opinions expressed in an Internet discussion group about terrorism become more extreme the longer the group is in existence."

Traps:

1. Do not give credit for answers that focus on other social psychological concepts such as conformity or peer pressure.
2. Do not give credit for the notion of forming *two* extreme views within a group or a "widening of the gap" phenomenon produced through interaction with others. Group polarization means the whole group moves in the same direction.
3. Do not give credit for a change in *behavior* unless a corresponding change of view, attitude, or opinion is also mentioned.

Point 4: James-Lange theory of emotion

The student **must** demonstrate, through description or example, that emotional experience is a two-step process whereby a physiological or behavioral response precedes the emotion. The student does not need to specify the original stimulus event.

Typical examples:

1. "I'm running from the mean dog, therefore I'm scared."
2. "I'm screaming, so I'm afraid."
3. "I'm crying, I must be sad."

Trap: Do not score the point if the student's answer implies that the physiological and emotional responses are happening simultaneously ("I'm crying and I am sad").

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Question 2 (cont'd.)

Point 5: CS—UCS

The student **must** use classical conditioning terms *or an example* to establish any one of the following:

1. the CS should come before the UCS, or
2. conditioning is weaker if the CS doesn't come before the UCS, or
3. conditioning is stronger with a short separation between the CS and the UCS, or
4. stronger conditioning occurs with repeated pairings of the CS and the UCS or if the CS and UCS are paired over time.

Typical examples:

1. "In Pavlov's experiment, the bell comes before the meat."
2. "Classical conditioning works best when the CS occurs shortly before the UCS."

Traps:

1. Do not score the point if a student uses the terms incorrectly in his or her example (e.g., do not score "In Pavlov's experiment, the UCS, a bell, is presented before the CS, food.")
2. Saying that the CS and UCS are "associated," "paired," or "correlated" does not establish that the CS comes before the UCS.

Point 6: Refractory period

The student must establish that it takes time before a neuron can ready itself to generate a second action potential ("fire again"). Note that:

1. if a student says the cell is "at rest" or "resting," he or she must differentiate between this rest and the cell's resting potential by noting that, during the refractory period, the cell is recharging, recovering, or unable to fire.
2. a student may use an analogy (e.g., "a neuron is like a camera flash" or "a neuron is like a toilet") to establish the point if the student establishes the notion of recharging.

Typical examples:

1. "The refractory period is the time it takes for a neuron to repolarize so it can fire again."
2. "A refractory period is like the time when a camera flash is recharging so it can be used again."

Trap: Do not score "The refractory period is when the neuron is at rest" because it does not establish the idea that the cell is readying to fire again.

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Question 2 (cont'd.)

Point 7: Sound localization

The student **must** both:

1. establish that sound localization requires input from both ears.
2. indicate that sound localization is possible because the sound waves reach the nearer ear before or in less time than they reach the more distant ear.

Typical examples:

1. "A sound to your left reaches your left ear slightly before it reaches your right ear, allowing us to determine the location of the sound."
2. "It is difficult to pinpoint a sound coming from directly in front or directly behind you because it reaches both ears at the same time."

Traps:

1. Do not score the point if the student refers to intensity or distance of a sound without noting the importance of a two-ear or binaural difference.
2. Do not score the point if the student refers to two ears without establishing the importance of a time differential.

Point 8: Spontaneous recovery

The student **must** establish both:

1. Reference to extinction, partial extinction, or the diminishing/decreasing of a response.
2. The recurrence of the response after a rest period. Do not score the point if the student is clearly confused about the stimulus that leads to the recovered response.

Typical examples:

1. "If Pavlov's dog's response to the bell was extinguished, it might return after a period of time passed."
2. "A pigeon in a Skinner box would stop pecking if it no longer received food. However, the next time it was placed in the box, it would begin pecking."

Trap:

1. Do not score the point if the student discusses reconditioning. ("The dog responded when Pavlov gave food again" or "the pigeon began pecking when the food reward was resumed.")
2. Do not score the point unless the student clearly establishes a rest period. "The response recurs later" does not score, but "the response comes back after a rest period" or "the response comes back the next day" are acceptable.